# ABHIJITH SIVAPRASADAN

*Project Engineer, Process | Energy & Process Engineering | Power Plant Process Design*
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## PROFILE

Energy and process-oriented engineer completing an M.Sc. in Sustainable Energy Engineering at KTH, with experience in industrial energy performance mapping, thermal-fluid modelling, measurement-system validation, technical documentation and Python-based engineering analysis. Strong fit for Wärtsilä Energy process project work: process calculations, process descriptions, design documentation, stakeholder coordination and decarbonisation-focused power plant engineering. Motivated to grow into PFD/P&ID, device-list and project-specific process design work for engine power plants, hybrid energy systems and balancing power solutions.

## CORE FIT FOR WÄRTSILÄ

**Process and energy engineering:** Thermal systems, process heat, utilities, industrial energy systems, sustainable power generation, engine power plant interest and process-improvement mindset.
**Process design documentation:** Process calculation and simulation mindset, PFD/P&ID and device-list fundamentals, process description writing, technical reports, assumptions tracking and design-package support.
**Decarbonisation and power systems:** District heating dispatch optimisation, CHP/heat-pump operation, energy storage and hybrid-system scenarios, cost/CO2 objectives, energy efficiency and operational KPIs.
**Data and project delivery:** Python, pandas, NumPy, SciPy, PuLP, MATLAB scripting basics, Excel/Sheets, reproducible workflows, stakeholder communication and multicultural teamwork.

## EXPERIENCE

**Test Engineer / Master Thesis Student**
*Siemens Energy AB, Fluid Dynamic Lab, Finspång* Apr 2025 - Nov 2025

- Developed steady-state compressible CFD and conjugate heat-transfer models in ANSYS Fluent for high-temperature gas-turbine test-rig hardware, translating boundary conditions, assumptions and physical constraints into engineering KPIs.
- Performed mesh/near-wall validation, sensitivity studies and simulation-to-test interpretation; documented model limitations and next-step experimental verification needs for structured R&D decision-making.
- Commissioned and verified NI-DAQ measurement chains with thermocouples, pressure sensors and LabVIEW workflows, building a practical mindset for instrumentation lists, signal verification and data reliability.
- Supported troubleshooting and technical documentation for rig work, including structured root-cause reasoning and communication of findings to industrial and academic stakeholders.

**Energy Efficiency Intern**
*Alleima AB, Sandviken (Remote)* Dec 2024 - Jun 2025

- Built an ISO 50001/EED-aligned energy performance mapping approach for industrial energy systems, including KPI/EnPI definitions, baselines, load drivers and normalisation logic.
- Assessed metering and data readiness across operational contexts and proposed measurement plans for reliable energy follow-up, deviation detection and improvement tracking.
- Developed analytical workflows to compare operating states, identify loss mechanisms and support improvement roadmaps for utilities, process heat, compressed air and heat recovery.

**Backend Engineer**
*QBurst, Thiruvananthapuram, India* Aug 2021 - Jul 2023

- Developed backend APIs, data workflows and automated test pipelines using Python, TypeScript, Postman, Git and CI-oriented development practices.
- Built transferable habits for process engineering work: structured debugging, reproducible calculations, version control, validation checks, technical documentation and collaborative delivery.

**Student Intern - Pyrolysis Research**
*KTH Royal Institute of Technology, Stockholm* Feb 2024 - Jun 2024

- Conducted cost-analysis research on oil extraction from polymers and reviewed plastic-pyrolysis reactor concepts to support early-stage process and technology assessment.

## RELEVANT PROJECTS

**Heating Dispatch Optimisation - District Heating System**
*KTH Course Project (Python, PuLP, MILP)* 2025

- Implemented multi-asset heat-supply dispatch optimisation with operational constraints, CHP and heat-pump operation, electricity-price coupling, CO2 cost, maintenance outages, storage and KPI reporting.
- Analysed technology mix, fuel use, seasonal operation and cost/emissions trade-offs - relevant to process calculations and decarbonisation-focused energy system design.

**Smart Energy Island and Hybrid Energy System Modelling**
*KTH Course Projects (IDA ICE, HOMER Pro, SAM)* 2024 - 2025

- Modelled heat demand, PV, batteries, heat pumps, solar thermal, TES and district heating scenarios; evaluated CAPEX, OPEX, LCOE, payback, autonomy and system performance.
- Developed scenario and KPI reporting for energy-service decision support, combining technical assumptions with system-level economic and operational outputs.

**Distribution Grid Study with EV, PV and Storage Integration**
*KTH Course Project (Power-flow / time-series simulation)* 2025

- Performed load-flow, N-1 contingency and 24-hour time-series analysis for a CIGRE MV network with EV charging, PV and storage scenarios; evaluated voltage stability, line loading and mitigation options.

## EDUCATION

**M.Sc. in Sustainable Energy Engineering (expected 2026)**
*KTH Royal Institute of Technology, Stockholm* 2023 - 2026
Relevant coursework: Sustainable Power Generation, Modelling of Energy Systems, Energy Management, Practical Optimization of Energy Networks, Energy Systems for Sustainable Development, Applied Heat and Power Technology, Numerical Heat Transfer.

**Circular Economy for Energy Storage**
*Aalto University / Unite! Virtual Exchange, Finland* Oct 2024 - Dec 2024

**B.Tech in Mechanical Engineering**
*APJ Abdul Kalam Technological University, Kerala, India* 2017 - 2021
Relevant foundation: thermodynamics, heat and mass transfer, fluid mechanics, thermal engineering, machine design, CAD/CAE and manufacturing fundamentals.

## TECHNICAL SKILLS

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|---|---|
| **Process design:** | Process calculations and simulations; PFD/P&ID and device-list fundamentals; process descriptions; design-package support; documentation of assumptions and requirements. |
| **Energy / power systems:** | Engine power plant interest; sustainable power generation; process heat and utilities; CHP/heat pumps; balancing power, hybrid energy systems, energy storage and decarbonisation. |
| **Analytics / optimisation:** | Python/PuLP MILP; scenario analysis; cost/CO2 objectives; KPI reporting; sensitivity analysis; data quality; operational-state comparison. |
| **Programming / tools:** | Python, pandas, NumPy, SciPy, matplotlib, PuLP, MATLAB scripting basics, Excel/Sheets, Git, Docker, Postman, REST APIs. |
| **Simulation / engineering:** | ANSYS Fluent/CHT, ANSYS Meshing, SpaceClaim, IDA ICE, HOMER Pro, SAM, Siemens NX, SolidWorks; thermal-fluid modelling and validation workflows. |
| **Testing / documentation:** | NI-DAQ, NI MAX, LabVIEW, thermocouples, pressure sensors, signal verification, measurement plans, technical reports and English documentation. |

## ADDITIONAL INFORMATION

Certifications: SSG Entre / Industrial Safety | Applied CFD - Siemens | IBM Python for Data Science | Digital Manufacturing & Design Technology | Schneider Electric sustainability course
Leadership: Founder of SAEINDIA Collegiate Club chapter; SAE team lead/captain and secretary/treasurer roles; Students' Nobel NightCap operations and safety role.
Languages: English - fluent/professional | Malayalam - native | Swedish - beginner, actively studying